

SE1 Module 3 Notes: Agile Methodologies

CS0025 Software Engineering 1. Built from CS0025-M3S0, M3S1, and M3S2. Read top to bottom; the blue boxes are memory aids I made (they are not from the slides).

How To Use These Notes

Module 3 has two chapters. **Chapter 1** is Agile in general and Scrum. **Chapter 2** is five other Agile methods: Crystal, DSDM, FDD, Lean (LSD), and XP. The exam likes lists, so the memory aids matter most for: the four values, the Scrum roles and events, the DSDM phases, the FDD practices, and the Lean principles.

Method	One-Line Identity	Hook To Remember
Agile	Iterative, customer-focused way of building software	Small steps, feedback, change is welcome
Scrum	Agile framework built on sprints and a few roles	Sprints, backlog, daily 15 minutes
Crystal	Agile that puts people first, by Alistair Cockburn	People over process
DSDM	Agile framework with fixed time and budget	Timebox, MoSCoW, prototype
FDD	Build it feature by feature	Feature is the unit of work
Lean (LSD)	Agile from Toyota lean manufacturing	Kill waste, just in time
XP	Short releases for changing requirements	Customer in the target

Chapter 1: Agile and Scrum

What Is Agile?

Agile is a group of approaches where requirements and solutions **evolve** through teamwork between **self-organizing, cross-functional teams** and their customers or end users. It pushes four things: **adaptive planning, evolutionary development, early delivery, and continual improvement**, and it welcomes quick, flexible response to change. **Scrum is one type of Agile.**

Agile breaks the work into **small increments** with little up-front planning and design. At the end of each iteration a **working product is shown to stakeholders**, which lowers risk and lets the product adapt quickly. Agile touches every function: planning, analysis, design, coding, testing.

The Four Agile Values

We Value This More	Over This	What It Really Means
Individuals and interactions	Processes and tools	Tools matter, but competent people working together matter more.
Working software	Comprehensive	Documentation helps, but the point of

	documentation	development is software, not paperwork.
Customer collaboration	Contract negotiation	A contract is no substitute for working closely with customers to find what they need.
Responding to change	Following a plan	A plan must not be so rigid that it cannot absorb new technology, priorities, or understanding.

Memory Aid

The four values, left side only: **People, Product, Partners, Pivot.**

People (individuals), Product (working software), Partners (customer collaboration), Pivot (responding to change).

Trap: Agile does not say the right side is worthless. It says the left side is valued **more**.

The Twelve Agile Principles

The slides give them in three groups of four. Learn them as the idea behind each one:

1. Customer satisfaction through **early and continuous delivery** of valuable software.
2. **Welcome changing requirements**, even late in development.
3. Deliver working software **frequently** (weeks rather than months).
4. **Close, daily cooperation** between business people and developers.
5. Build projects around **motivated individuals** and trust them.
6. **Face-to-face conversation** is the best communication (co-location).
7. **Working software** is the primary measure of progress.
8. **Sustainable development**: keep a constant pace.
9. Continuous attention to **technical excellence** and good design.
10. **Simplicity**: the art of maximizing the work not done.
11. Best architectures, requirements, and designs come from **self-organizing teams**.
12. Regularly **reflect** on how to become more effective, then adjust.

Memory Aid

Chunk the twelve into three stories of four:

Customer story (1 to 4): satisfy, welcome change, deliver often, talk daily.

Team story (5 to 8): motivated people, face to face, software is progress, steady pace.

Craft story (9 to 12): excellence, simplicity, self-organizing, reflect and adjust.

Agile Vs Waterfall

Waterfall flows **sequentially** from start to end. Agile is **incremental and iterative**. The customer sees a waterfall product only at the end, but sees an Agile product early and often.

Point	Agile	Waterfall
Flow	Incremental and iterative	Sequential, start to end

Structure	Broken into individual models designers work on	Not broken into individual models
Customer sees product	Early and frequently, can make changes	Only at the end of the project
Structure feel	Considered unstructured compared to waterfall	More secure because it is plan oriented
Project size	Small projects done quickly; large projects are hard to estimate	All sorts of projects can be estimated and completed
Errors	Can be fixed in the middle of the project	Whole product tested only at the end; a requirement error can force a restart
Iteration	Short iterations of 2 to 4 weeks, very little planning	Big phases; each phase ends with a detailed description of the next
Documentation	Less priority than software	Top priority, even used to train staff or hand over to another team
Testing	Every iteration has its own testing phase, so regression testing each release	Only after the development phase
Delivery	Shippable features at the end of each iteration	All features delivered at once after a long implementation
Testers and developers	Work together; user acceptance at the end of every sprint	Work separately; user acceptance at the end of the project
Developers in planning	Close communication; developers help analyze requirements	Developers not involved in requirements and planning; delays between tests and coding

Memory Aid

Waterfall = one big **waterfall**, you only see the lake (finished product) at the bottom.

Agile = many small **agile** steps, you see something after every step.

Quick test: if the statement says "only at the end" or "top priority documentation", it is Waterfall.

What Is Scrum?

Scrum is an **agile process framework for managing complex knowledge work**, first aimed at software development, research, and advanced technologies. It is designed for **teams of ten or fewer**, who split work into goals finished in **timeboxed iterations called sprints**. A sprint is **no longer than a month** and most commonly **two weeks**. Progress is tracked and re-planned in **15-minute timeboxed daily meetings called daily scrums**.

Regular project management builds the whole product in one pass. Scrum delivers **several iterations** of a product so stakeholders get the **highest business value in the least time**. It asks for frequent planning and goal setting, which keeps the team focused and productive.

Benefits Of Scrum (Seven)

- Flexibility and adaptability
- Creativity and innovation
- Lower costs
- Quality improvement
- Organizational synergy
- Employee satisfaction
- Customer satisfaction

Memory Aid

Seven benefits, in a 2 + 2 + 3 pattern:

Ways of working: Flexibility, Creativity.

Business results: Lower costs, Quality.

Who is happy, inside out: Organization (synergy), Employees, Customers.

The Three Scrum Roles

Role	What They Do	Key Words
Scrum Master	Facilitator of the scrum process. Holds the daily meetings, makes sure scrum rules are enforced, coaches and motivates the team, removes impediments , and gives the team the best conditions to deliver.	Facilitator, removes impediments, enforces the rules
Product Owner	Represents the stakeholders (usually customers). Sets product expectations, records changes, and administers the backlog , a constantly updated to-do list. Prioritizes goals for each sprint by value to stakeholders.	Stakeholders, backlog, prioritizes
Scrum Development Team	A self-organized group of three to ten people with the business, design, analytical, and development skills to do the work. They self-administer tasks and are jointly responsible for each sprint goal.	3 to 10, self-organized, does the work

Memory Aid

Master = Manages the process, not the product. The Scrum Master never decides what to build.

Owner = Owns the list. The Product Owner owns and orders the backlog.

Team = Three to ten doers. (Some slides say teams of ten or fewer overall; the team itself is 3 to 10.)

The Six Scrum Events

Order on the slide: Sprint, Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective, Backlog Refinement.

Memory Aid

Sentence: Some **P**andas **D**ance, **R**est, **R**elax, **R**epeat.

= Sprint, Planning, Daily, Review, Retrospective, Refinement.

Event	What Happens	Time For A 2-Week Sprint
Sprint	Basic unit of development. A timebox with a length fixed in advance, between one week and one month (two weeks most common).	2 weeks typical
Sprint Planning	Team agrees on scope, selects backlog items that fit one sprint, prepares the sprint backlog , and agrees on a sprint goal (a short description of what they forecast to deliver). Tasks are forecast, usually by voting.	4 hours
Daily Scrum	Starts precisely on time even if members are missing; same time and place every day. Anyone may attend, but only the development team contributes.	15 minutes
Sprint Review	Team reviews completed and not-completed work, presents completed work to stakeholders, and collaborates on what to do next. Incomplete work cannot be demonstrated.	2 hours
Sprint Retrospective	Team reflects on the past sprint and agrees on improvement actions. Facilitated by the Scrum Master.	1.5 hours
Backlog Refinement	Ongoing; formerly called grooming . Reviews backlog items so they are clear and ready for future sprints. Items may be split, acceptance criteria clarified, dependencies found.	Ongoing

Memory Aid

Times for a two-week sprint: **4 - 15 - 2 - 1.5**.

Planning **4 hours** (longest, it sets the whole sprint), Daily **15 minutes**, Review **2 hours**, Retro **1.5 hours**.

Review and Retro are the two "look back" meetings and Retro is the shorter one.

Trap: Review is **with stakeholders** and shows **finished work only**. Retro is **the team alone**, about the **process**.

The Three Daily Scrum Questions

1. What did I complete **yesterday** that helped the team meet the sprint goal?
2. What do I plan to complete **today** toward the sprint goal?

3. Do I see any **impediment** that could block me or the team?

Memory Aid

Y-T-B: Yesterday, Today, Blockers.

The Three Retrospective Questions

1. What **went well** during the sprint?
2. What **did not go well**?
3. What **could be improved** for better productivity next sprint?

Memory Aid

Well, Not Well, Better. (Keep, Stop, Improve in plain words.)

Sprint Planning Checklist

- Mutual discussion and agreement on the scope of work
- Select product backlog items that can be completed in one sprint
- Prepare a sprint backlog with the work needed
- Agree on a sprint goal
- Four hours for a two-week sprint
- Forecast tasks, usually by voting

Chapter 2: Other Agile Methods

Crystal

Crystal is an Agile approach that focuses on **people and their interactions** rather than processes and tools. It was developed by **Alistair Cockburn**, who believed people's skills, talents, and the way they communicate have the biggest impact on a project's outcome.

Two beliefs: teams can **streamline their own processes** as they work and become more optimized, and **projects are unique and dynamic**, so they need specific methods.

Phase	What Happens
1. Chartering	Create the development team, do a preliminary feasibility analysis, develop an initial plan, fine-tune the methodology.
2. Cyclic Delivery	Two or more delivery cycles: update and refine the release plan; implement a subset of requirements through one or more program-test-integrate iterations; deliver the integrated product to real users; review the project plan and methodology.
3. Wrap Up	Deployment into the user environment, post-deployment reviews and reflections.

Memory Aid

Crystal phases: **Charter, Cycle, Wrap**. Start a club (charter), run the meetings in cycles, wrap up.

Crystal = Cockburn = people. Both start with a hard C sound.

DSDM (Dynamic Systems Development Method)

DSDM is an Agile **project delivery framework**. Users must be **actively involved**, teams are **given the power to make decisions**, and the focus is **frequent delivery** of the product.

Three Techniques

Technique	Meaning
Timeboxing	Split the project into portions, each with a fixed budget and delivery date . Requirements are prioritized per portion. Because time and budget are fixed, the only variable is the requirements : if time or money runs short, the lowest-priority requirements are dropped .
MoSCoW	A prioritization technique to agree with stakeholders how important each requirement is. Must have, Should have, Could have, Won't have .
Prototyping	Build prototypes early so shortcomings are found early and future users can test-drive the system. Gives good user involvement , a key DSDM success factor.

Memory Aid

MoSCoW is an acronym with filler letters. The capitals carry the meaning: **M**ust, **S**hould, **C**ould, **W**on't.

Priority falls in that order. Won't means not this time.
Timeboxing in one line: **time and money are fixed, scope bends.**

The Seven DSDM Phases

1. Pre-project
2. Feasibility Study
3. Business Study
4. Functional Model Iteration
5. Design and Build Iteration
6. Implementation
7. Post-project

Memory Aid

Sentence: **Pam Finds Big Fat Dogs In Parks.**

= **P**re-project, **F**easibility, **B**usiness study, **F**unctional model, **D**esign and build, **I**mplementation, **P**ost-project.

Shape: two bookends (Pre and Post), two studies in front (Feasibility, Business), two iterations in the middle (Functional, Design and Build), then Implementation.

FDD (Feature Driven Development)

FDD is built around **designing and building features**. Unlike other Agile methods it describes very **specific, short phases of work done separately per feature**. The slide names these activities: domain walkthrough, design inspection, promote to build, code inspection, and design.

The Eight Practices

1. Domain Object Modeling
2. Development by Feature
3. Component or Class Ownership
4. Feature Teams
5. Inspections
6. Configuration Management
7. Regular Builds
8. Visibility of Progress and Results

Memory Aid

Chunk as **1 + 3 + 3 + 1: Model it, Own it, Check it, Show it.**

Model it: Domain Object Modeling.

Own it (three): Development by Feature, Class Ownership, Feature Teams.

Check it (three): Inspections, Configuration Management, Regular Builds.

Show it: Visibility of progress and results.

Lean Software Development (LSD)

Lean translates **lean manufacturing** principles to software. It is adapted from the **Toyota Production System** and built on "**Just in time production**". It aims to **increase speed and decrease cost**.

The Seven Lean Principles

#	Principle (Two Names)	Meaning
1	Eliminating Waste	If an activity can be skipped, or the result reached without it, it is waste. Examples: abandoned partial code, paperwork, extra features customers rarely use.
2	Amplifying Learning	Development is continuous learning through iterations. Software value is measured in fitness for use , not conformance to requirements.
3	Defer Commitment (Decide as late as possible)	Uncertainty is normal, so delay decisions until they can be based on facts, not assumptions .
4	Early Delivery (Deliver as fast as possible)	The sooner a defect-free product ships, the sooner feedback comes. "Not the biggest that survives, but the fastest."
5	Empowering the Team	Managers learn to listen to developers. Follows the Agile principle: build around motivated people and trust them.
6	Building Integrity	The customer has a whole-system experience: how it is advertised, delivered, deployed, accessed, how intuitive it is, its price, how well it solves the problem.
7	Optimize the Whole	Defects pile up as work is split into parts, so find and remove root causes. Systems are the product of their interactions , not just the sum of parts. "Think big, act small, fail fast; learn rapidly."

Memory Aid

Sentence: **Even A Dull Day Earns Bright Optimism.**

= **E**liminate waste, **A**mplify learning, **D**efer commitment, **D**eliver early, **E**mpower team, **B**uild integrity, **O**ptimize whole.

Number 1 is the most fundamental: **Eliminate waste.**

Seven Wastes Of Software Development (Cisco Version)

The DevNet module (see the NetComms notes) names the seven wastes. Useful if a question asks "which is a waste":

- Partially done work
- Extra processes
- Extra features
- Task switching
- Waiting

- Motion
- Defects

Memory Aid

Please **E**at **E**very **T**asty **W**affle, **M**y **D**arling = Partial work, Extra processes, Extra features, Task switching, Waiting, Motion, Defects.

XP (Extreme Programming)

XP is very helpful when **requirements change constantly** or the customer is **unsure about the functionality**. It pushes **frequent releases in short development cycles**, which improves productivity and gives checkpoints where customer requirements are easy to add. **XP develops software keeping the customer in the target.**

The Six XP Phases

Phase	Activities
Planning	Identify stakeholders and sponsors; infrastructure requirements; security information; Service Level Agreements and conditions
Analysis	Capture stories in the " parking lot "; prioritize them; scrub stories for estimation; define the time iteration; plan resources for development and QA
Design	Break down tasks; prepare test scenarios for each task; regression automation framework
Execution	Coding; unit testing; manual test scenarios; defect reports; convert manual to automated regression tests; mid-iteration review; end-of-iteration review
Wrapping	Small releases; regression testing; demos and reviews; new stories as needed; process improvements from the end-of-iteration review
Closure	Pilot launch; training; production launch; SLA guarantee assurance; review SOA strategy; production support

Memory Aid

Phases: **P-A-D-E-W-C** = **P**lan **A**nd **D**esign, **E**xecute, **W**rap, **C**lose.

Parking lot = where user stories wait to be prioritized (Analysis).

Story Cardboard = sticky notes on a board (manual, slower). Online Storyboard = the same stories kept in an online tool that several teams can share.

One-Page Cram Sheet

Topic	Remember
Agile values	People, Product, Partners, Pivot (left side valued more)
Scrum roles	Scrum Master (process, removes blocks), Product Owner (backlog, stakeholders), Dev Team (3 to 10)
Scrum events	Sprint, Planning, Daily, Review, Retro, Refinement
Times (2-week sprint)	Planning 4 h, Daily 15 min, Review 2 h, Retro 1.5 h
Daily questions	Yesterday, Today, Blockers
Retro questions	Well, Not well, Better
Crystal	Cockburn; Charter, Cycle, Wrap
DSDM	Timebox, MoSCoW, Prototype; 7 phases PFBFDIP
FDD	8 practices: Model, Own (3), Check (3), Show
Lean	7 principles: Eliminate waste first; Toyota; just in time
XP	Changing requirements; 6 phases PADEWC; parking lot